Developer Guide

This section is for people working on the SWAT+ Fortran source: building from a clone, running the test suite, and submitting changes back to `swat-model/swatplus`.

If you only need to run a prebuilt SWAT+ executable, see Install instead.

Pages

- Build with CMake. The base out-of-source build flow that every platform uses. Covers configuration, compiler selection (gfortran, ifort, ifx), debug vs release, and installation.
- Visual Studio on Windows. End-to-end workflow for Visual Studio 2022 with the Intel oneAPI Fortran compiler. Cloning, configuring, debugging against an input dataset.
- VS Code Codespaces. Running a SWAT+ build and debug session inside a GitHub Codespaces virtual instance. Useful when you do not want to install a Fortran toolchain locally.
- Testing. The scenario regression test suite that compares model output against golden datasets in `data/`. Driven by `ctest`.
- Tagging and releases. How the build system pulls the version from `git describe` and projects it into `main.f90` and the executable name. Covers the release tag flow.
- Coding conventions. Fortran 90/95 style rules for SWAT+ source contributions: portability, readability, robustness, arrays, I/O, and obsolescent features to avoid.
- Contributing. Fork, branch, pull request, and the CI checks that run on every PR.
- Code of conduct. The Contributor Covenant, applied to the SWAT+ project community.